BUSINESS REQUIREMENTS • APPROVAL PACK

Vehicle Shipment Allocation & Lead Tracking

AURA Dealership Management System · GT Automotive / CAM Motors

Document version	1.0	**Date**	9 October 2026
Status	Draft — pending business approval	**Business owner**	To be nominated
Source	User-supplied “Functional Requirements Document: Vehicle Shipment Allocation & Lead Tracking Flow”		

> **Approval purpose:** agree the business scope, allocation rules, communications, exceptions and acceptance criteria before engineering begins.
> **No implementation is authorized by this draft.** Recommendations below are proposals, not approved business rules. All blocking decisions in Section 11 must be resolved before implementation.

Approval pack contents

Business case · scope · current-system constraints · lifecycle · functional requirements · data and imports · communications · permissions · controls · exception handling · decision register · acceptance tests · rollout and sign-off.

Prepared from the supplied requirements and a targeted review of existing AURA lifecycle and reservation constraints. This is not a complete technical design, delivery estimate, or confirmation of production data readiness.

1. Business need and intended outcomes

Dealership staff need to connect customer demand to expected vehicle shipments before physical VINs are available, keep customers informed when arrival estimates change, and assign incoming VINs fairly without mismatching specifications or allocating the same capacity twice.

- Prevent “Vehicle Allocated” without an active, compatible shipment allocation.
- Support both spreadsheet imports and manual shipment creation.
- Maintain a reliable baseline ETA and a separately visible current ETA.
- Permit controlled shipment reassignment when customer preferences or logistics change.
- Randomly match VINs within exact specification groups and report every exception.
- Keep dealership ownership, existing financial controls and physical inventory integrity intact.

Success measure	Proposed acceptance target
Invalid advancement / double allocation	Zero accepted transitions without eligible capacity; zero duplicate active VIN claims in UAT.
Traceability	Every allocation, reassignment, ETA revision and VIN assignment has actor, timestamp and before/after history.
Communications	Every required event has a traceable notification outcome; failed or missing-contact cases remain visible, never silently marked sent.
Matching accuracy	100% of confirmed automatic assignments match all approved specification fields.
Efficiency	Operations to establish the current manual baseline and approve a processing-time target before go-live.

2. Scope and boundaries

In scope for the proposed first release

- Dealer-scoped shipment register, detail view, specification lines, quantities, month/year ETA, status and audit history.
- CSV and Excel templates, validation preview, explicit import confirmation and row-error reports.
- Assignment of eligible pre-booked customer demand to compatible shipment capacity.
- Hard “Vehicle Allocated” gate, capacity checks, reassignment and release workflows.
- Allocation and ETA-change customer emails using dealership branding and approved wording.
- VIN manifest upload, randomized exact-match allocation, assignment summary and unmatched queues.
- Lead/customer visibility, operational reports, notification status and role-based controls.
- Reconciliation with existing inventory, reservation holds and deal commitment.

Out of scope unless separately approved

Carrier/GPS tracking, automated customs clearance, procurement or supplier-payment redesign, changes to vehicle prices or tax calculations, automatic refunds, lender decision changes, guaranteed handover dates, cross-dealer stock transfers, WhatsApp/SMS notifications, and direct ERP posting changes. Shipment arrival does not itself complete PDI, registration, settlement or customer delivery.

Not an exclusion: compatibility with existing bookings, logistics, holds, deals and delivery workflows is mandatory. Any required interface work must be scoped before approval of the technical design.

3. Stakeholders and existing-system constraints

Role	Responsibility
Business sponsor / dealership management	Approve scope, eligibility, fairness policy, exceptions and release readiness.
Sales manager / advisors	Confirm customer specifications, assign capacity within permission, communicate changes and manage exceptions.
Logistics / inventory team	Create shipments, maintain ETAs and manifests, reconcile expected versus received units.
Finance	Approve the payment/hold boundary and cancellation/reassignment implications.
Customer communications / policy owner	Approve templates, disclaimers, recipient rules and sending policy.
Engineering / QA	Validate integration design, imports, concurrency, recovery, security and UAT evidence.

> **Material alignment issue:** the current lead phase model is new → contacted → qualified → proposal → negotiation → won/lost. “Pre-Book” and “Vehicle Allocated” must not silently replace that model. Existing vehicle interests describe specifications; physical claims also involve paid reservation holds and deal commitment. The requested manifest-driven VIN assignment therefore needs an explicit decision about when it becomes a binding physical claim.

Recommended design direction, pending approval: keep a separate fulfilment/allocation status alongside the sales phase. Shipment allocation reserves expected quantity, not a physical VIN. When VINs arrive, matching proposes or creates a controlled claim through the existing allocation authority; it must never overwrite another customer's valid hold. Final VIN commitment must remain consistent with the agreed finance/deal rules.

4. Definitions and lifecycle

Term	Business meaning
Shipment	A dealership-owned incoming batch with a unique reference, expected arrival month/year and specification quantities.
Shipment line	A normalized make/model/year/variant/exterior colour/interior colour configuration and expected quantity.
Pre-Book	Customer demand eligible for shipment assignment. Exact payment/document prerequisites require decision D1.
Assigned shipment	A valid link between a customer demand line and a matching shipment line; capacity consumption starts at the approved point.
Vehicle Allocated	Expected shipment capacity reserved for the customer; does not by itself mean a VIN is assigned or handover is ready.

VIN assigned	A specific vehicle identity linked through the approved physical-claim process, with a unique active claimant.
Baseline / current ETA	Original communicated arrival month/year versus the latest revised estimate. Neither guarantees delivery.

Proposed customer fulfilment flow: Eligible Pre-Book → Shipment Assigned → Vehicle Allocated → VIN Pending / VIN Matched → VIN Confirmed → Existing delivery workflow. Reassignment creates a new allocation version; it does not delete the previous one.

Proposed shipment flow: Draft → Open for Allocation → In Transit → Arrived → Closed; Cancelled is an exception path. Only approved active statuses accept new allocation. Partial receipts leave outstanding quantity open. Closing or cancelling cannot strand live customer allocations without an exception plan.

5. Functional requirements and business rules

ID	Requirement	Observable outcome
BR-01	Create and maintain shipments through UI or CSV/XLSX import.	Shipment reference, dealer, specification lines, quantities, ETA month/year, status and history visible together.
BR-02	Validate incoming shipment data before committing.	Unknown specifications, duplicate references, invalid dates and non-positive/non-integer quantities are identified by row; no hidden partial import.
BR-03	Assign only eligible demand to compatible active shipment lines.	System verifies dealer, specification, quantity, pre-book eligibility and remaining capacity.
BR-04	Enforce the advancement gate on every write path.	UI, API, imports and automation cannot mark Vehicle Allocated without the same valid allocation criteria.
BR-05	Freeze baseline allocation details upon advancement.	Original specification, shipment, quantity and ETA remain auditable; later changes are versioned.
BR-06	Send initial allocation email.	One event per successful allocation version includes vehicle details, quantity, reference, estimated arrival and disclaimer.
BR-07	Permit controlled reassignment and preference changes.	Target eligibility and capacity are revalidated. Target claim and old release succeed together; failure leaves the original allocation intact.
BR-08	Record ETA revisions and notify affected assigned customers.	Old/new month-year, reason, actor, affected-recipient count and batch progress are visible. Saving the same ETA is not a date-shift event.
BR-09	Import shipment VIN manifests with reconciliation.	VINs associate to the correct dealer, shipment and specification line; duplicates and incompatible rows are rejected or held for review.
BR-10	Randomly match within exact specification groups.	Only eligible unassigned demand and claim-free matching VINs participate. No substitution of trim or colour to clear an exception.
BR-11	Produce complete matching results.	Successful assignments, unmatched customers, unassigned VINs and rejection reasons reconcile to all inputs.
BR-12	Protect established inventory and financial commitments.	Existing reservations, sales and committed VINs cannot be overwritten; any authorized change uses the existing release/refund gates.

BR-13	Support quantity-based demand and partial arrivals.	Expected, allocated, manifested, matched, received and released quantities are distinguishable; partial fulfilment follows the approved rule.
BR-14	Provide customer/lead and operations visibility.	Authorized staff see shipment reference, current/baseline ETA, allocation history, VIN status and email outcomes without switching dealerships.

Capacity and matching controls

- Proposed capacity formula: allocatable quantity = approved expected quantity − active allocated quantity. VIN receipt does not increase expected capacity a second time.
- Concurrent requests for the last unit must produce at most one successful claim; the other request receives a clear conflict.
- Quantity reduction below active commitments must be blocked or routed into an approved shortage-resolution workflow; never silently remove customers.
- Canonical specifications must include model, variant/trim and exterior/interior colours. Make and model year are recommended additional exact-match fields; required fields cannot treat blank as a wildcard.
- Normalize accepted aliases during reviewed import. Do not apply fuzzy substitution during allocation.
- Random selection operates only inside an exact-match group. Record the run, eligible population, algorithm version and reproducible selection evidence. A retry must reuse the confirmed result, not reshuffle existing winners.
- Shortage priority, partial quantity allocation and manual overrides require approval in D4–D6. Randomness must not implicitly change customer entitlement.

6. Data, imports and reconciliation

Record	Minimum business data
Shipment	Dealer, unique shipment reference, status, ETA month/year, optional supplier/vessel reference, notes, revision history.
Shipment specification line	Make/model, model year if applicable, variant, exterior/interior colour, expected quantity, allocated quantity and manifest/receipt totals.
Customer allocation	Lead/customer and relevant deal/demand line, shipment line, quantity, allocation status, baseline/current ETA, specification snapshot, actor, timestamps and release/reassignment reason.
VIN manifest / match run	Import reference, shipment, VIN, canonical specifications, validation result, existing inventory link, eligibility snapshot, run result and confirmation actor.
Notification event	Allocation/ETA revision, dealer, recipient reference, template version, queued/sent/failed state, attempt count and provider outcome.

Shipment template columns: shipment_reference, eta_month, eta_year, make, model, model_year, variant, exterior_color, interior_color, quantity; optional supplier_reference and notes. Final required/optional specification fields depend on D3.

VIN template columns: shipment_reference, VIN, make, model, model_year, variant, exterior_color, interior_color. The selected dealership comes from the authorized session, not a trusted spreadsheet field.

Upload → parse and normalize → validate → preview errors and changes → explicit confirm → atomic apply → downloadable result. Recommended first-release policy: reject the import if any row is invalid; the reviewer fixes and re-uploads it. Re-upload must not duplicate shipment lines, inventory, customer claims or emails. Duplicate VIN checks must include existing dealer inventory and active claims, not merely the current file.

Import limits and supported workbook layouts will be agreed during technical design. Reject macro execution, unsafe formulas and unsupported file types; exports must neutralize spreadsheet formula injection. Original uploads and results follow the approved retention/access policy.

7. Customer communications

Event	Recipients and timing	Required content
Vehicle Allocated	Customer contact attached to the successful allocation; queued immediately after the transaction commits.	Dealer identity/contact, customer name, specification/quantity, allocation reference, baseline estimated arrival month/year, disclaimer.
ETA changed	All currently assigned customers affected by that shipment revision, including those awaiting VINs. Exclude released/cancelled allocations; revalidate relevance before sending.	Previous and revised ETA, shipment/allocation reference, change explanation, contact details and disclaimer.
Reassignment	Recommended additional allocation-update email; approval required in D7.	New shipment, changed specifications/ETA, reference and disclaimer.

> **Proposed disclaimer — policy owner approval required:**
> “Your estimated shipment arrival is [Month Year]. This is an estimate, not a guaranteed vehicle delivery or handover date, and may change due to shipping, port, customs, transit or other logistics factors. We will notify you of updates. Your dealership will confirm handover separately once the vehicle is ready and all required checks are complete.”

Draft email copy

Allocation subject: [Dealership] — Your vehicle shipment allocation
Dear [Customer], we have allocated [Quantity] × [Vehicle specification] to shipment [Reference]. Estimated shipment arrival: [Month Year]. Your allocation reference is [Allocation reference]. [Approved disclaimer] For assistance, contact [Dealer contact].

ETA update subject: [Dealership] — Updated estimated arrival for your vehicle
Dear [Customer], the estimated arrival for shipment [Reference], linked to your [Vehicle specification], has changed from [Old month/year] to [New month/year]. [Approved reason, where appropriate.] [Approved disclaimer] For assistance, contact [Dealer contact].

Use the dealership's configured outbound sender; never fall back to another dealer. Missing email or sending configuration creates a visible exception without inventing delivery success. Retries reuse the original event identity. Once already sent, a corrected ETA generates a new revision rather than altering history. Queued superseded revisions should be marked superseded so customers are not sent stale updates. “Sent” means provider accepted; delivered/bounced is shown only when evidence is available. No real customer emails during UAT.

8. Permissions, audit and operational reports

Proposed role matrix below must map to existing granular permissions; role names alone must not imply access.

Action	Advisor	Sales manager	Logistics	Finance / audit
View eligible leads / allocations	Assigned scope	Authorized dealer scope	Operational scope	Relevant read scope
Create shipments / import manifests / revise ETA	No	By explicit grant	By explicit grant	Read only
Assign / reassign before VIN commitment	By explicit grant	Yes, scoped	No by default	Read only
Confirm VIN run / resolve overrides	No	Approval authority	Prepare / propose	Approve financial impacts
Change a committed VIN / refund	No	Existing approval flow only	No unilateral change	Existing approval flow only

Every read, write, import, job, export and notification is dealer-scoped. Branch visibility follows existing scope where applicable. Audit records retain actor/system origin, dealership, timestamp in UTC, before/after values, reason, source file/run reference and notification linkage. Display times use dealership timezone; month/year ETAs are calendar periods, not converted timestamps.

Reports: shipment capacity by specification; assigned customers with baseline/current ETA; overallocated/shortage exceptions; unmatched customers/unassigned VINs; import failures; notification failures; reassignment and override history. Filters include dealer, permitted branch, shipment, ETA period, model, colour and status. Customer data and financial details respect existing access rules in exports as well as screens.

9. Exceptions and non-functional controls

Situation	Required handling
Shipment delayed/cancelled, quantity reduced or split	Show affected customers, preserve history, require an authorized resolution; no silent reassignment or duplicate capacity.
Customer changes preference / withdraws	Revalidate target specifications and capacity; preserve financial holds until the applicable cancellation/refund rules permit release.
Insufficient or excess VINs	Allocate only per approved fairness/quantity policy; report remainder without cross-spec substitution.
VIN already held, sold, committed or duplicated	Exclude and report; do not override existing ownership.
Failure during import, matching or reassignment	No partial inconsistent claims; restart/retry safely using the same operation identity.
Email failure	Allocation remains valid; retry and alert staff. Do not roll back customer allocation or falsely show sent.
Lead converted, merged or closed while work is queued	Revalidate current customer/deal linkage and eligibility; do not duplicate allocations or notify a superseded contact.

Proposed service criteria: ordinary screens should respond within 2 seconds at the 95th percentile under an agreed test workload; larger imports and communication batches should run asynchronously with progress and retry visibility. Confirm peak shipment sizes, recipient counts and concurrent user targets before engineering estimates. Retention, recovery objectives and monitoring thresholds remain approval items, not assumed guarantees.

10. Source-to-requirement traceability

Supplied FRD item	BRD coverage
3.1 File and interactive shipment creation	BR-01–02; Sections 4 and 6
3.2 Pre-Book advancement and reassignment	BR-03–05, BR-07, BR-12–13; D1–D3
3.3 Allocation and ETA-shift email/disclaimer	BR-06, BR-08; Section 7; D7
3.4 VIN upload, randomized matching and audit	BR-09–11; Sections 5–6 and 8; D4–D6
Summary Rules 1–4	BR-04, BR-08, BR-07, BR-10 respectively

11. Approval decision register

All decisions below are **OPEN**. Approvers may accept the recommendation or record an alternative. D1–D8 block implementation scope; D9–D10 must close before final estimation/go-live planning.

ID / owner	Decision required	Recommended position for approval
D1 · Sales + Finance	What qualifies as Pre-Book? Is payment required? Does shipment assignment immediately consume capacity?	Separate fulfilment status from lead phase. Require an approved pre-book record and confirmed specifications. Consume capacity only at Vehicle Allocated; define payment/document prerequisites explicitly, without inventing a deposit amount.
D2 · Finance + Sales + Engineering	Does manifest matching create a permanent VIN claim before deal commitment? How does it coexist with paid holds?	Matching proposes candidates; confirm through the existing claim authority. Preserve existing holds and final commitment rules. If early binding VIN allocation is required, explicitly approve that lifecycle change.
D3 · Inventory + Sales	Exact matching fields and valid specification catalogue?	Make/model/year/variant/exterior/interior colour where applicable. Explicit “not applicable” for genuinely absent specifications; no blank wildcards.
D4 · Management	Who wins when matching demand exceeds VIN supply?	Oldest eligible allocation first for entitlement; randomly assign VIN identities within that eligible group. The source's alternative of random customer selection during shortages requires explicit approval.
D5 · Sales + Finance	Multi-unit customers, split shipments and partial fulfilment?	Track each requested quantity line. Default all-or-nothing fulfilment per line unless customer-approved partial fulfilment is recorded. Confirm whether shipment splitting is included in release one.

D6 · Management + Logistics	Automatic commit versus preview/approval; manual VIN overrides?	Automatic randomized proposal after valid import, authorized confirmation before final claims. Any override needs reason, same-spec validation and approval; no bypass of existing financial holds.
D7 · Communications owner	Email audience/timing, legal copy and reassignment notification?	Initial allocation and every effective ETA revision to active assigned customers; also send reassignment update. Approve templates and exception follow-up ownership. Email only for release one.
D8 · Logistics + Management	Definition of active shipment; shortages/cancellation/closure authority?	Open and In Transit eligible; Arrived only with reconciled remaining capacity. No cancellation/closure with unresolved active allocations.
D9 · Operations + Engineering	Historical data migration, retention and workload?	Pilot one selected dealer/shipment; validate existing commitments first. Import history without triggering customer mail unless separately authorized. Agree file/batch limits and retention before estimates.
D10 · Sponsor	Named approvers, budget, delivery priority and release date?	Nominate accountable owners and request technical estimate only after D1–D8 close. No delivery date or cost commitment in this draft.

12. Acceptance criteria and UAT checklist

Business users and QA should retain pass/fail evidence against each scenario. Policy-dependent cases use the approved decisions above.

ID	Given / When / Then
UAT-01	Given a valid UI entry or import, when confirmed, then one dealer-scoped shipment with correct specification quantities and ETA is created.
UAT-02	Given an invalid row or duplicate reference, when import is previewed, then errors identify the row/field and no invalid data is committed.
UAT-03	Given no shipment, inactive shipment, wrong specifications or insufficient capacity, when Vehicle Allocated is requested through any channel, then advancement is rejected with a reason.
UAT-04	Given one remaining unit and two simultaneous requests, when both attempt allocation, then only one succeeds and capacity never becomes negative.
UAT-05	Given an eligible allocation, when advancement completes, then baseline details are preserved and exactly one initial email event is queued.
UAT-06	Given an ETA change, when saved, then current ETA updates, baseline remains, and every affected active customer has one traceable notification outcome. Re-saving the same date produces none.
UAT-07	Given reassignment, when target validation succeeds, then capacity moves atomically with history; when it fails, the original allocation remains untouched.
UAT-08	Given exact-match demand and valid manifest VINs, when the approved matching process runs, then each confirmed VIN has one claimant and all specification fields match.
UAT-09	Given a shortfall, excess VINs or multi-unit demand, when matching runs, then approved D4/D5 rules apply and every unmatched remainder is reported.

UAT-10	Given a retry or repeated upload, when processing resumes, then existing assignments and notifications are not duplicated or randomly reshuffled.
UAT-11	Given a held/sold/committed VIN, when imported or selected, then the conflict is reported without changing its existing claimant.
UAT-12	Given another dealership's record or unauthorized role, when viewed, changed or exported, then access is denied without exposing customer data.
UAT-13	Given an email failure, missing address or absent dealer sender, when dispatch runs, then staff see the exception, allocation remains valid, and sent is not falsely reported.
UAT-14	Given cancellation, quantity reduction or customer withdrawal, when processed, then affected commitments remain auditable and existing financial release gates still apply.
UAT-15	Given a matched vehicle, when a user attempts handover, then existing PDI, settlement, registration and delivery controls are still enforced.
UAT-16	Given a historical migration or test run, when processed, then no real customer communication is sent without separate explicit authorization.
UAT-17	Given rapid ETA revisions or customer reassignment while mail is queued, when dispatch occurs, then superseded events do not send stale shipment information.

13. Readiness, delivery sequence and change control

1. **Business review:** resolve decisions, nominate owners, approve scope and communications.
2. **Technical design:** map existing pre-book, logistics, inventory and allocation flows; define one authoritative claim mechanism, schema/migrations, API contracts, import templates and failure recovery.
3. **Design review:** agree shipment list/detail, assignment/reassignment dialog, VIN review, exception queues and customer-status presentation.
4. **Implementation after authorization:** shipment/import foundation → allocation gate/reassignment → communications → VIN matching/reporting. This is sequencing only, not a committed schedule.
5. **UAT:** dealer isolation, concurrent claims, retries, malformed files, historical records and regression of finance/delivery controls.
6. **Pilot:** selected dealership and shipment, reconciled opening quantities, trained operations team and monitored email batches.
7. **Release:** signed UAT, approved templates, tested migration/recovery, assigned support owner and documented rollback/feature-disable plan. Disabling new work must not erase existing allocations or sent messages.

Go/no-go checklist: D1–D10 resolved; business and technical owners signed; no unresolved critical defects; capacity/VIN reconciliation balanced; permission matrix verified; notifications tested without customer sends; historical migration reviewed; support alerts and retry ownership established.

After approval, changes to eligibility, VIN commitment, fairness, communications or scope require a new document revision with business impact and renewed approval. Approval of this BRD does not itself authorize production migration, customer messaging or publishing.

14. Formal sign-off

Approver	Approval responsibility	Name / decision / date
Business sponsor	Scope, value, priority and funding	Pending
Sales lead	Eligibility, customer workflow, reassignment and fairness	Pending
Logistics / inventory lead	Shipment lifecycle, imports, quantities and VIN reconciliation	Pending
Finance lead	Reservation, commitment, cancellation and refund boundaries	Pending
Communications / policy owner	Recipients, templates and disclaimer	Pending
Engineering / QA lead	Feasibility, integration scope, estimate and acceptance coverage	Pending

Approval options: Approved with all decisions recorded / Approved subject to listed conditions / Revise and resubmit. Record conditions, excluded scope and the exact approved version. No signatures or approvals have been assumed.

Revision history

Version	Date	Change
1.0	9 October 2026	Initial approval draft from supplied FRD; added lifecycle alignment, decision register, controls, traceability and UAT.